Homework 3

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Homework 3 - Tidyverse

Conceptual Questions

1. If your working directory is myfolder/homework/, what relative path would you specify to get the file located at myfolder/MyData.csv?
    1. "../MyData.csv"
2. What are the major benefits of using R projects?
    1. I can create a project organization directly in RStudio, set the working directory, and manage git and github directly in RStudio (I love this!).
3. What is Git and what is GitHub?
    1. Git is a version control software that is run locally on my computer. It tracks changes to files and allows me to make those changes permanent or revert back to a previous version.
    2. GitHub is a cloud-based platform that hosts git repositories. It is a remote backup of repositories. It allows collaboration between team members by tracking changes and allowing each team member to upload their changes and others to pull those changes to their local environment. It also tracks versions and will alert the team member if there are merge conflicts allowing the team member to decide how to handle the conflicts.
4. What are the two main differences between a tibble and a data.frame?
    1. If one column is called from a tibble the tibble allows the column to remain a tibble instead of a vector.
    2. The tibble displays the data in a nicer format by only printing the data that is visible in the output. It also lists the number of columns and rows along with all the column names and types.
    3. Tibbles do not automatically convert character variables to factors.

Reading Delimited Data

Glass Data

Locate the glass data and read the comma separated values files into a tibble using the tidyverse function read_csv, but first load the library to load the functions in this session.

```
library(readr)
#?read_csv
glass_data <- read_csv("https://www4.stat.ncsu.edu/online/datasets/glass.data", col_names=c(
```

```
Rows: 214 Columns: 11
-- Column specification --------------------------------------------------------
Delimiter: ","
dbl (11): Id, RI, Na, Mg, Al, Si, K, Ca, Ba, Fe, Type

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
glass_data
```

```
# A tibble: 214 x 11
      Id    RI    Na    Mg    Al    Si     K    Ca    Ba    Fe  Type
   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
 1     1  1.52  13.6  4.49  1.1   71.8  0.06  8.75     0  0        1
 2     2  1.52  13.9  3.6   1.36  72.7  0.48  7.83     0  0        1
 3     3  1.52  13.5  3.55  1.54  73.0  0.39  7.78     0  0        1
 4     4  1.52  13.2  3.69  1.29  72.6  0.57  8.22     0  0        1
 5     5  1.52  13.3  3.62  1.24  73.1  0.55  8.07     0  0        1
 6     6  1.52  12.8  3.61  1.62  73.0  0.64  8.07     0  0.26     1
 7     7  1.52  13.3  3.6   1.14  73.1  0.58  8.17     0  0        1
 8     8  1.52  13.2  3.61  1.05  73.2  0.57  8.24     0  0        1
 9     9  1.52  14.0  3.58  1.37  72.1  0.56  8.3      0  0        1
10    10  1.52  13    3.6   1.36  73.0  0.57  8.4      0  0.11     1
# i 204 more rows
```

Yeast Data

Identify the correct number spaces for the start of each column to read the fixed width data into a tibble using the read_delim function since the data is fixed width not csv. Count the spaces for the start of each column ADT1_YEAST 0.58 0.61 0.47 0.13 0.50 0.00 0.48 0.22 MIT

```
#?read_fwf
yeast_data <- read_fwf('https://www4.stat.ncsu.edu/online/datasets/yeast.data', fwf_widths(c
```

```
Rows: 1484 Columns: 10
-- Column specification --------------------------------------------------------

chr (2): Sequence, class
dbl (8): mcg, gvh, alm, mit, erl, pox, vac, nuc

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
yeast_data
```

```
# A tibble: 1,484 x 10
   Sequence     mcg   gvh   alm   mit   erl   pox   vac   nuc class
   <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <chr>
 1 ADT1_YEAST  0.58  0.61  0.47  0.13   0.5   0    0.48  0.22 MIT
 2 ADT2_YEAST  0.43  0.67  0.48  0.27   0.5   0    0.53  0.22 MIT
 3 ADT3_YEAST  0.64  0.62  0.49  0.15   0.5   0    0.53  0.22 MIT
 4 AAR2_YEAST  0.58  0.44  0.57  0.13   0.5   0    0.54  0.22 NUC
 5 AATM_YEAST  0.42  0.44  0.48  0.54   0.5   0    0.48  0.22 MIT
 6 AATC_YEAST  0.51  0.4   0.56  0.17   0.5   0.5  0.49  0.22 CYT
 7 ABC1_YEAST  0.5   0.54  0.48  0.65   0.5   0    0.53  0.22 MIT
 8 BAF1_YEAST  0.48  0.45  0.59  0.2    0.5   0    0.58  0.34 NUC
 9 ABF2_YEAST  0.55  0.5   0.66  0.36   0.5   0    0.49  0.22 MIT
10 ABP1_YEAST  0.4   0.39  0.6   0.15   0.5   0    0.58  0.3  CYT
# i 1,474 more rows
```

Excel Data

Set my working directory to be the root directory so it will render and run code from the same place.

```
#install.packages("here")
library(here)
```

```
Warning: package 'here' was built under R version 4.5.3
```

```
here() starts at C:/Users/cbhut/OneDrive/Documents/GitHub/Data-Science-ED-
Group
```

Download the wine file and align the working directory and the directory the file is rendering from so I can indicate the correct location of the file for both.

```r
library(readxl)
#?read_excel

getwd()
```

```
[1] "C:/Users/cbhut/OneDrive/Documents/GitHub/Data-Science-ED-Group/HW_3"
```

```r
here()
```

```
[1] "C:/Users/cbhut/OneDrive/Documents/GitHub/Data-Science-ED-Group"
```

```r
list.files()
```

```
[1] "Homework3_CH.qmd"        "Homework3_CH.rmarkdown" "HW3_CM.qmd"
[4] "HW3_ES.qmd"              "white-wine.xlsx"
```

```r
file.exists("white-wine.xlsx")
```

```
[1] TRUE
```

```r
file.exists("HW_3/white-wine.xlsx")
```

```
[1] FALSE
```

Open, load and render the wine file by specifying the location so I can access the file through the qmd file and when the file is rendered.

```r
white_wine_data <- read_excel(here("HW_3", "white-wine.xlsx"))
white_wine_data
```

```
# A tibble: 4,898 x 12
   `fixed acidity` `volatile acidity` `citric acid` `residual sugar` chlorides
             <dbl>              <dbl>         <dbl>            <dbl>     <dbl>
 1             7                 0.27          0.36             20.7     0.045
 2             6.3               0.3           0.34              1.6     0.049
 3             8.1               0.28          0.4               6.9     0.05
 4             7.2               0.23          0.32              8.5     0.058
 5             7.2               0.23          0.32              8.5     0.058
 6             8.1               0.28          0.4               6.9     0.05
 7             6.2               0.32          0.16              7       0.045
 8             7                 0.27          0.36             20.7     0.045
 9             6.3               0.3           0.34              1.6     0.049
10             8.1               0.22          0.43              1.5     0.044
# i 4,888 more rows
# i 7 more variables: `free sulfur dioxide` <dbl>,
#   `total sulfur dioxide` <dbl>, density <dbl>, pH <dbl>, sulphates <dbl>,
#   alcohol <dbl>, quality <dbl>
```

Combine Excel Files

White Wine

Read in the column names from the second sheet in the same white wine Excel file.

```r
wine_names <- read_excel(here("HW_3", "white-wine.xlsx"), sheet=2)
wine_names
```

```
# A tibble: 12 x 1
   Variables
   <chr>
 1 fixed_acidity
 2 volatile_acidity
 3 citric_acid
 4 residual_sugar
 5 chlorides
 6 free_sulfur_dioxide
 7 total_sulfur_dioxide
 8 density
 9 pH
10 sulphates
11 alcohol
12 quality
```

Add the column names to the white wine data file

```r
colnames(white_wine_data) <- wine_names$Variables
white_wine_data
```

```
# A tibble: 4,898 x 12
   fixed_acidity volatile_acidity citric_acid residual_sugar chlorides
           <dbl>            <dbl>       <dbl>          <dbl>     <dbl>
 1           7               0.27        0.36           20.7     0.045
 2           6.3             0.3         0.34            1.6     0.049
 3           8.1             0.28        0.4             6.9     0.05
 4           7.2             0.23        0.32            8.5     0.058
 5           7.2             0.23        0.32            8.5     0.058
 6           8.1             0.28        0.4             6.9     0.05
 7           6.2             0.32        0.16            7       0.045
 8           7               0.27        0.36           20.7     0.045
 9           6.3             0.3         0.34            1.6     0.049
10           8.1             0.22        0.43            1.5     0.044
# i 4,888 more rows
# i 7 more variables: free_sulfur_dioxide <dbl>, total_sulfur_dioxide <dbl>,
#   density <dbl>, pH <dbl>, sulphates <dbl>, alcohol <dbl>, quality <dbl>
```

Add a new column that indicates all of the observations in the tibble are white wines.

```r
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
white_wine_data <- white_wine_data |>
  mutate(type = 'White')
white_wine_data
```

```
# A tibble: 4,898 x 13
   fixed_acidity volatile_acidity citric_acid residual_sugar chlorides
           <dbl>            <dbl>       <dbl>          <dbl>     <dbl>
 1           7               0.27        0.36           20.7     0.045
 2           6.3             0.3         0.34            1.6     0.049
 3           8.1             0.28        0.4             6.9     0.05
 4           7.2             0.23        0.32            8.5     0.058
 5           7.2             0.23        0.32            8.5     0.058
 6           8.1             0.28        0.4             6.9     0.05
 7           6.2             0.32        0.16            7       0.045
 8           7               0.27        0.36           20.7     0.045
 9           6.3             0.3         0.34            1.6     0.049
10           8.1             0.22        0.43            1.5     0.044
# i 4,888 more rows
# i 8 more variables: free_sulfur_dioxide <dbl>, total_sulfur_dioxide <dbl>,
#   density <dbl>, pH <dbl>, sulphates <dbl>, alcohol <dbl>, quality <dbl>,
#   type <chr>
```

Red Wine

Read in the semicolon delimited data into a red wine file.

```
red_wine <- read_delim("https://www4.stat.ncsu.edu/~online/datasets/red-wine.csv", delim=';')
```

```
Rows: 1599 Columns: 12
-- Column specification --------------------------------------------------------
Delimiter: ";"
dbl (12): fixed acidity, volatile acidity, citric acid, residual sugar, chlo...

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
red_wine
```

```
# A tibble: 1,599 x 12
   `fixed acidity` `volatile acidity` `citric acid` `residual sugar` chlorides
             <dbl>              <dbl>         <dbl>            <dbl>     <dbl>
 1             7.4               0.7           0                 1.9     0.076
 2             7.8               0.88          0                 2.6     0.098
 3             7.8               0.76          0.04              2.3     0.092
 4            11.2               0.28          0.56              1.9     0.075
```

```
 5              7.4                 0.7            0                  1.9      0.076
 6              7.4                 0.66           0                  1.8      0.075
 7              7.9                 0.6            0.06               1.6      0.069
 8              7.3                 0.65           0                  1.2      0.065
 9              7.8                 0.58           0.02               2        0.073
10              7.5                 0.5            0.36               6.1      0.071
# i 1,589 more rows
# i 7 more variables: `free sulfur dioxide` <dbl>,
#   `total sulfur dioxide` <dbl>, density <dbl>, pH <dbl>, sulphates <dbl>,
#   alcohol <dbl>, quality <dbl>
```

Rename the columns for the red wine to match the columns from the white wine.

```r
colnames(red_wine) <- wine_names$Variables
red_wine
```

```
# A tibble: 1,599 x 12
   fixed_acidity volatile_acidity citric_acid residual_sugar chlorides
           <dbl>            <dbl>       <dbl>          <dbl>     <dbl>
 1           7.4             0.7         0               1.9     0.076
 2           7.8             0.88        0               2.6     0.098
 3           7.8             0.76        0.04            2.3     0.092
 4          11.2             0.28        0.56            1.9     0.075
 5           7.4             0.7         0               1.9     0.076
 6           7.4             0.66        0               1.8     0.075
 7           7.9             0.6         0.06            1.6     0.069
 8           7.3             0.65        0               1.2     0.065
 9           7.8             0.58        0.02            2       0.073
10           7.5             0.5         0.36            6.1     0.071
# i 1,589 more rows
# i 7 more variables: free_sulfur_dioxide <dbl>, total_sulfur_dioxide <dbl>,
#   density <dbl>, pH <dbl>, sulphates <dbl>, alcohol <dbl>, quality <dbl>
```

Add a new 'type' column and label each observation in the tibble as red.

```r
red_wine <- red_wine |>
  mutate(type = 'red')
red_wine
```

```
# A tibble: 1,599 x 13
   fixed_acidity volatile_acidity citric_acid residual_sugar chlorides
```

```
          <dbl>            <dbl>       <dbl>          <dbl>     <dbl>
 1          7.4             0.7         0               1.9       0.076
 2          7.8             0.88        0               2.6       0.098
 3          7.8             0.76        0.04            2.3       0.092
 4         11.2             0.28        0.56            1.9       0.075
 5          7.4             0.7         0               1.9       0.076
 6          7.4             0.66        0               1.8       0.075
 7          7.9             0.6         0.06            1.6       0.069
 8          7.3             0.65        0               1.2       0.065
 9          7.8             0.58        0.02            2         0.073
10          7.5             0.5         0.36            6.1       0.071
# i 1,589 more rows
# i 8 more variables: free_sulfur_dioxide <dbl>, total_sulfur_dioxide <dbl>,
#   density <dbl>, pH <dbl>, sulphates <dbl>, alcohol <dbl>, quality <dbl>,
#   type <chr>
```

Combine the white wine and red wine files to one file.

```r
?bind_rows
```

```
starting httpd help server ... done
```

```r
wine_data <- bind_rows(white_wine_data, red_wine)
wine_data
```

```
# A tibble: 6,497 x 13
   fixed_acidity volatile_acidity citric_acid residual_sugar chlorides
           <dbl>            <dbl>       <dbl>          <dbl>     <dbl>
 1           7               0.27        0.36           20.7     0.045
 2           6.3             0.3         0.34            1.6     0.049
 3           8.1             0.28        0.4             6.9     0.05
 4           7.2             0.23        0.32            8.5     0.058
 5           7.2             0.23        0.32            8.5     0.058
 6           8.1             0.28        0.4             6.9     0.05
 7           6.2             0.32        0.16            7       0.045
 8           7               0.27        0.36           20.7     0.045
 9           6.3             0.3         0.34            1.6     0.049
10           8.1             0.22        0.43            1.5     0.044
# i 6,487 more rows
# i 8 more variables: free_sulfur_dioxide <dbl>, total_sulfur_dioxide <dbl>,
#   density <dbl>, pH <dbl>, sulphates <dbl>, alcohol <dbl>, quality <dbl>,
#   type <chr>
```